INCH-POUND

MIL-S-19500/276B
28 April 1989
SUPERSEDING
MIL-S-19500/276A
29 January 1968

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS), SILICON TYPES 2N2323, 2N2324, 2N2326, 2N2328, 2N2329 AND S VERSIONS 2N2323A, 2N2324A, 2N2326A, 2N2328A, AND S VERSIONS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNPN silicon, reverse-blocking-triode thyristors. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-5).

1.3 Maximum ratings.

Type	V_{RM}	V_{RM} (non-rep)	V_{FBXM}	I_O 1/	I_{FSM} 2/	V_{KGM}	T_{op}	T_{STG}	Barometric pressure
	V (pk)	V (pk)	V (pk)	A dc	a	V (pk)	°C	°C	mmHg
2N2323,S	50	75	50 3/	0.22	15	6	-65 to +125	-65 to +150	---
2N2323A,S	50	75	50 4/	0.22	15	6	-65 to +125	-65 to +150	---
2N2324,S	100	150	100 3/	0.22	15	6	-65 to +125	-65 to +150	---
2N2324A,S	100	150	100 4/	0.22	15	6	-65 to +125	-65 to +150	---
2N2326,S	200	300	200 3/	0.22	15	6	-65 to +125	-65 to +150	---
2N2326A,S	200	300	200 4/	0.22	15	6	-65 to +125	-65 to +150	---
2N2328,S	300	400	300 3/	0.22	15	6	-65 to +125	-65 to +150	15
2N2328A,S	300	400	300 4/	0.22	15	6	-65 to +125	-65 to +150	15
2N2329,S	400	500	400 3/	0.22	15	6	-65 to +125	-65 to +150	15

1/ This average forward current is for an ambient temperature of 80°C and 180 electrical degrees of conduction. For other operating conditions see figure 3.
2/ Surge current is nonrecurrent. The rate of rise of peak surge current shall not exceed 40 A during the first 5 µs after switching from the "off" (blocking) to the "on" (conducting) state. This is measured from the point where the thyristor voltage has decayed to 90 percent of its initial blocking value.
3/ Gate connected to cathode through 1,000 ohm resistor.
4/ Gate connected to cathode through 2,000 ohm resistor.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, Dayton, OH 45444-5276 or by the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics (at 25°C, unless otherwise specified).

	V_{FM} I_{FM} = 40(pK) t_p = 8.5 ms, max	I_{HOX} 1/ 2/	V_{GT} 1/ 2N2323 through 2N2329 2N2323S through 2N2329S T_A = -65°C thru +125°C	I_{GT} 1/ 2N2323 through 2N2329 2N2323S through 2N2329S T_A = -65°C thru +125°C	V_{GT} 2/ 2N2323A through 2N2328A 2N2323AS through 2N2328AS T_A = -65°C thru +125°C	I_{GT} 2/ 2N2323A through 2N2328A 2N2323AS through 2N2328AS T_A = -65°C thru +125°C
	V	mA dc	V dc	µA dc	V dc	µA dc
Min			0.1		0.1	
Max	2.2	2.0	1.0	350	0.8	75

1/ Gate connected to cathode through 1,000 ohm resistor (for non-A types).
2/ Gate connected to cathode through 2,000 ohm resistor (for A-types).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

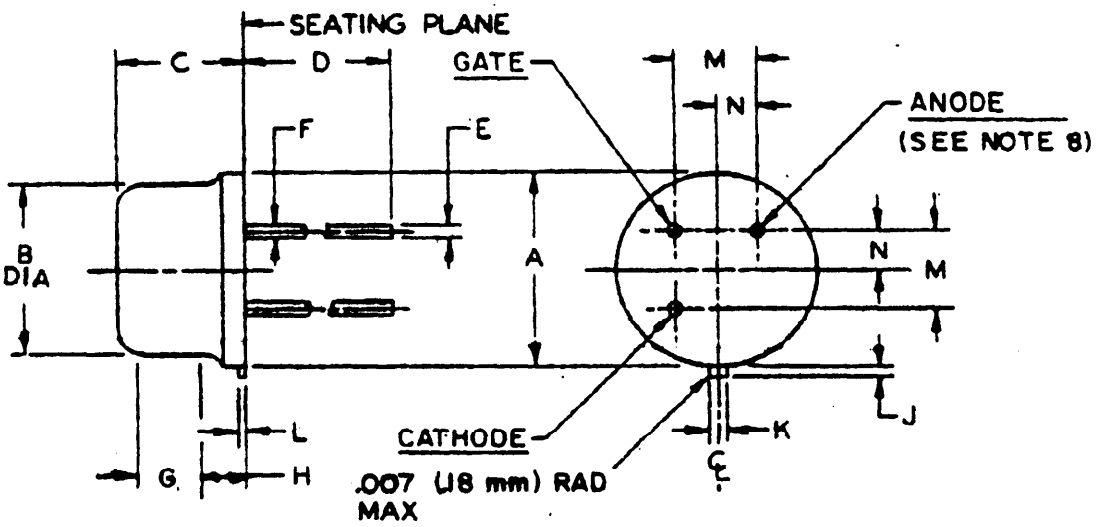
MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Dimensions					
	Inches		Millimeters		
Letter	Min	Max	Min	Max	Notes
A	.335	370	8.51	9.40	
B	.305	.335	7.75	8.51	
C	.240	.260	6.10	6.60	
D	See notes 9, 10, and 11				
E	.016	.021	0.41	0.53	3, 10
F	.016	.019	0.41	0.48	4, 10
G	.100	---	2.54	---	5
H	---	---	---	---	6
J	.029	.045	0.74	1.14	9
K	.028	.034	0.71	0.86	
L	.009	.125	0.23	3.18	
M	.1414 Nom		3.592 Nom		7
N	.0707 Nom		1.796 Nom		7

FIGURE 1. Physical dimensions.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
4. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
5. Variations on dimension B in this zone shall not exceed .010 (.25 mm).
6. Outline in this zone is not controlled.
7. When measured in a gauging plane .054 +.001, -.000 (1.37 +0.03, -0.00) below the seating plane of the thyristor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 preferred measured method.
8. The anode shall be internally connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All three leads.
11. For thyristor types 2N2323S, 2N2324S, 2N2326S, 2N2328S, 2N2329S, 2N2323AS, 2N2324AS, 2N2326AS, 2N2328AS, D is .500 (12.70 mm) minimum, and .750 (19.95 mm) maximum.
12. For thyristor types 2N2323, 2N2324, 2N2326, 2N2328, 2N2329, 2N2323, 2N2324, 2N2326A, 2N2328A, D is 1.500 (38.10 mm) minimum, and 1.750 (44.45 mm) maximum.

FIGURE 1. Physical dimensions - Continued.

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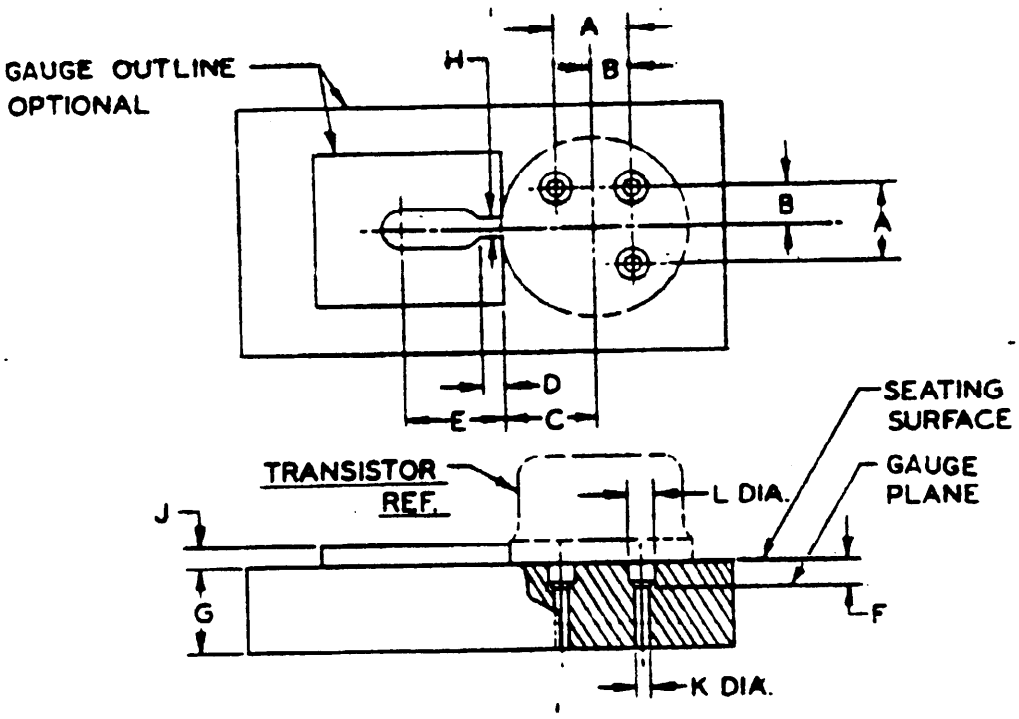


FIGURE 2. Gauge for lead and tab location.

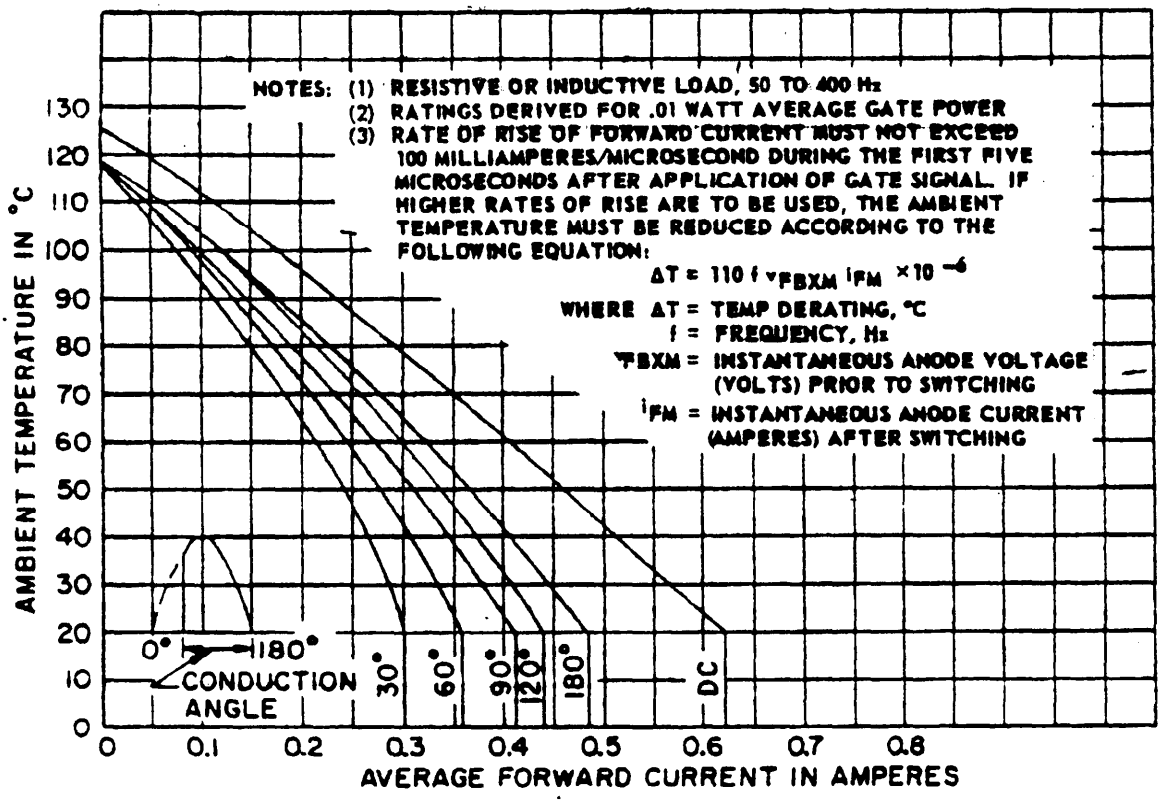
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Dimensions				
	Inches		Millimeters	
Letter	Min	Max	Min	Max
A	.1409	.1419	3.578	3.604
B	.0702	.0712	1.783	1.809
C	.182	.199	4.62	5.05
D	.009	.011	0.23	0.28
E	.125 Nom		3.18 Nom	
F	.054	.055	1.37	1.40
G	.372	.378	9.45	9.60
H	.0350	.0355	0.889	0.901
J	.150 Nom		3.81 Nom	
K	.0325	.0335	0.826	0.851
L	.0595	.0605	1.511	1.537

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedure shall be used: The use of a pin straightener prior to insertion in the gauge is permissible. The device being measured shall be inserted until its seating plane is .125 ±.010 (3.18 ±0.25 mm) from the seating surface of the gauge. A spacer may be used to obtain the .125 (3.18 mm) distance from the gauge seat prior to force application. A force of 8 ounces ±.5 ounce shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be sealed against the gauge.
4. The location of the tab locator, within the limits of dimension C, will be determined by the tab and flange dimension of the device being checked.

FIGURE 2. Guage for lead and tab location - Continued.

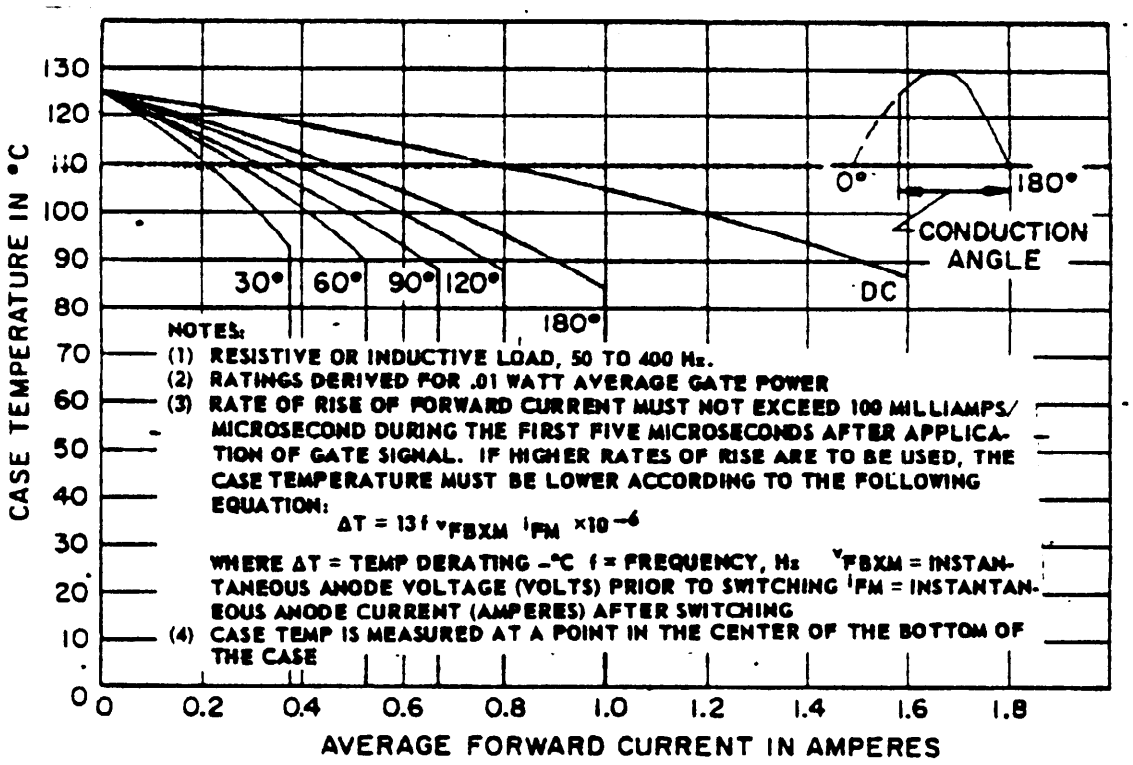
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Operating conditions (average forward current verses ambient temperature).

FIGURE 3. Operating conditions.

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Operating conditions (average forward current verses case temperature).

FIGURE 3. Operating conditions - Continued.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

I_{FBX}	DC forward blocking current with specified conditions applied to the gate.
I_{HOX}	DC holding current with specified conditions applied to the gate.
I_{RBX}	DC reverse blocking current with specified conditions applied to the gate.
R_{GK}	External resistance between gate and cathode terminals.
V_{AA}	Anode supply voltage (dc).
V_{FBX}	Forward blocking voltage, gate connected to the specified reference terminal.
V_{GG}	Gate supply voltage (dc).

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figure 1 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar or Alloy 52. Lead finish shall be gold plated, tin plated or solder dipped. Where a choice of lead material or finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500, except at the option of the manufacturer, the following marking may be omitted from the body of the device:

a. Country of origin.

b. Manufacturer's identification.

c. Polarity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein, except that the lot formation period requirements shall be six months maximum.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurements	
	JANS level	JANTX and JANTXV levels
9	I_{RBX1}, I_{FBX1}, V_{GT1}, I_{GT1}	Not applicable
10	See 4.3.1	Not applicable
11	I_{RBX1}, I_{FBX1}, V_{GT1}, I_{GT1}	I_{RBX1}, I_{FBX1}, V_{GT1}, I_{GT1}
12	See 4.3.2	See 4.3.1
13	Subgroups 2 and 3 of table I herein. ΔI_{RBX1} = +4.0 µA dc; ΔI_{FBX1} = +4.0 µA dc	Subgroup 2 of table I herein. ΔI_{RBX1} = +4.0 µA dc; ΔI_{FBX1} = +4.0 µA dc

4.3.1 Power burn-in. Power burn-in conditions are as follows:

T_A = 125°C, t = 96 hours (min)

V_{FBX}	R_{GK} = 1 kΩ	R_{GK} = 2 kΩ
50 V dc	2N2323, S	2N2323A, AS
100 V dc	2N2324, S	2N2324A, AS
200 V dc	2N2326, S	2N2326A, AS
300 V dc	2N2328, S	2N2328A, AS
400 V dc	2N2329, S	

All devices which turn "on" during the burn-in test shall be removed from the lot and the quantity removed shall be noted on the lot record.

4.3.2 Power burn-in for JANS. Power burn-in conditions for JANS are as follows:

t = 240 hours (min)
T_A = 75°C +5°C; I_Q = 220 mA dc; See figure 4.
R_{GK} = 1 kΩ for 2N2323 through 2N2329 and 2N2323S through 2N2329S.
R_{GK} = 2 kΩ for 2N2323A through 2N2328A and 2N2323AS through 2N2328AS.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein, except that the lot formation period requirements shall be six months maximum.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table IV herein.

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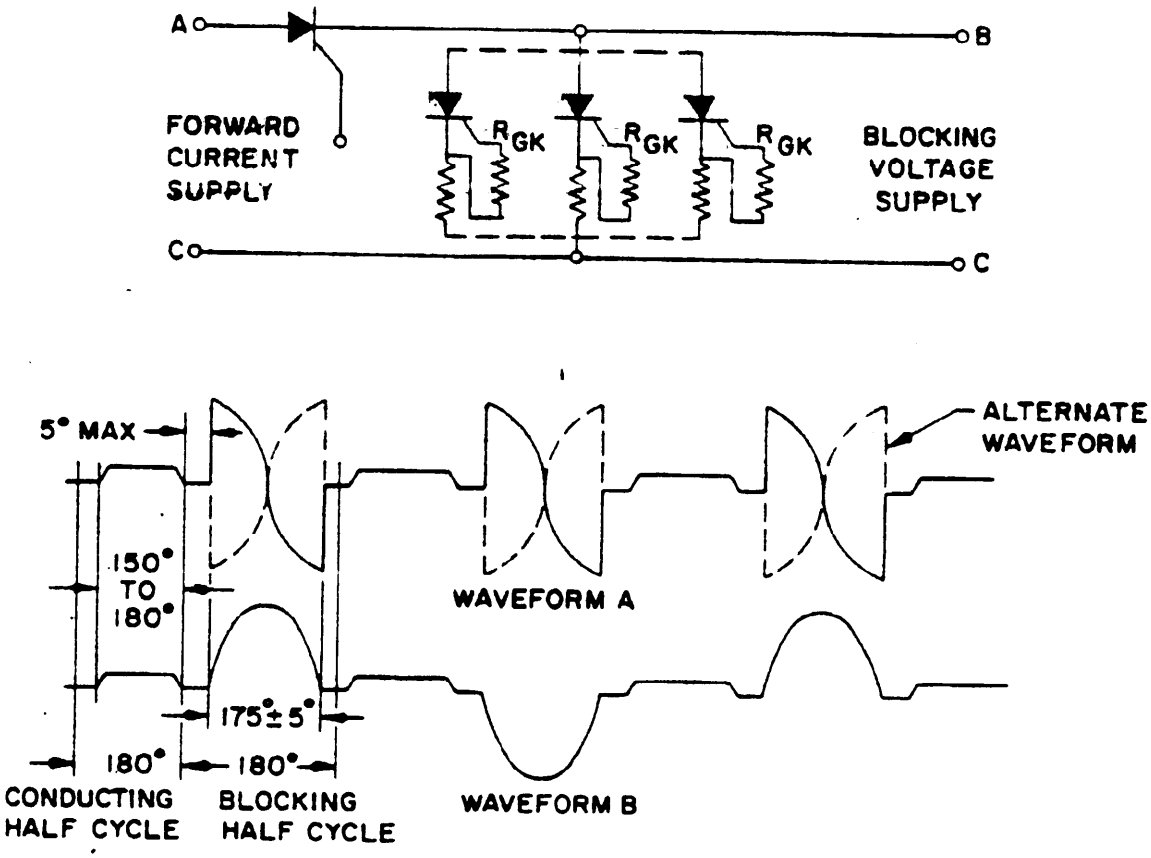


FIGURE 4. Test circuit and waveforms for intermittent operation life test.

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Test circuit details: The test circuit used must provide one of the waveforms shown. When a large number of devices are to be tested, they may be connected in parallel or series combinations with suitable provisions (resistive, reactive, etcs, methods) for proper division of forward current or blocking voltages. Suitable protective elements should be used to isolate defective devices without interrupting the test and to protect the remaining devices on test. The forward current supply shall consist of a low-voltage transformer with a means of adjusting the secondary voltage. It shall supply 22 mA dc average forward current to each device under test. The gate supply shall be in phase with the forward current supply. The gate average power input shall not be allowed to exceed .01 watt per device.

Operating conditions: The devices shall be subjected to the following operational conditions:

a. Power sources are to be 60 Hz sinusoidal waveform sources.
b. The conduction angle of I_0 shall be 150° to 180°.
c. The rated forward or reverse blocking voltage shall be started during the nonconducting half cycle no later than 5° after conduction has ceased. For waveform A, the 1RC time constant of the blocking voltage shall be 200 ±100 microseconds.
d. For accelerated steady-state operation, blocking voltage shall be equal to or greater than 30 V.
e. The duration of the blocking voltages shall be 175 ±5°.
f. The duration of life test shall be 1,000 (+72, -24) hours.
g. Ambient test temperature shall be 75 ±5°C.

FIGURE 4. Test circuit and waveforms for intermittent operation life test - Continued.

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FIGURE 1. Physical dimensions of transistor types 2N43A and 2N44A.

TABLE I. Group A inspection.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 1							
Visual and mechanical inspection	2071						
Subgroup 2							
Reverse blocking current	4211	DC method, condition B		I_{RBX1}	---	10	µA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 kΩ					
2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_2 = 2 kΩ					
2N2323, S, A, AS		V_R = 50 V dc					
2N2324, S, A, AS		V_R = 100 V dc					
2N2326, S, A, AS		V_R = 200 V dc					
2N2328, S, A, AS		V_R = 300 V dc					
2N2329, S		V_R = 400 V dc					
Forward blocking current	4206	DC method, condition B		I_{FBX1}	---	10	µA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 kΩ					
2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_2 = 2 kΩ					
2N2323, S, A, AS		V_{FBX} = 50 V dc					
2N2324, S, A, AS		V_{FBX} = 100 V dc					
2N2326, S, A, AS		V_{FBX} = 200 V dc					
2N2328, S, A, AS		V_{FBX} = 300 V dc					
2N2329, S		V_{FBX} = 400 V dc					
Reverse gate current	4219	V_{KG} = 6 V dc		I_{KG}	---	200	µA dc
Gate trigger voltage and current	4221	$V_2 = V_{FBX}$ = 6 V dc; R_L = 100Ω					
2N2323 thru 2N2329 and 2N2323S thru 2N2329S		R_e = 1 kΩ		V_{GT1} I_{GT1}	0.35 ---	0.80 200	V dc µA dc
2N2323A thru 2N2328A and 2N2323AS thru 2N2328AS		R_e = 2 kΩ		V_{GT1} I_{GT1}	0.35 ---	0.60 20	V dc µA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 3							
High temperature operation:		T_A = 125°C					
Reverse blocking current	4211	DC method, condition B		I_{RBX2}	---	100	µA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 kΩ					
2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_2 = 2 kΩ					
2N2323, S, A, AS		V_R = 50 V dc					
2N2324, S, A, AS		V_R = 100 V dc					
2N2326, S, A, AS		V_R = 200 V dc					
2N2328, S, A, AS		V_R = 300 V dc					
2N2329, S		V_R = 400 V dc					
Forward blocking current	4206	DC method, condition B		I_{FBX2}	---	100	µA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 kΩ					
2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_2 = 2 kΩ					
2N2323, S, A, AS		V_{FBX} = 50 V dc					
2N2324, S, A, AS		V_{FBX} = 100 V dc					
2N2326, S, A, AS		V_{FBX} = 200 V dc					
2N2328, S, A, AS		V_{FBX} = 300 V dc					
2N2329, S		V_{FBX} = 400 V dc					
Gate trigger voltage	4221			V_{GT3}	0.1	---	V dc
2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS		R_L = R_e = 1 kΩ					
2N2328, S, A, AS 2N2329, S		R_L = 4 kΩ, R_e = 2 kΩ					
2N2323, S, A, AS		V_2 = V_{FBX} = 50 V dc					
2N2324, S, A, AS		V_2 = V_{FBX} = 100 V dc					
2N2326, S, A, AS		V_2 = V_{FBX} = 200 V dc					
2N2328, S, A, AS		V_2 = V_{FBX} = 300 V dc					
2N2329, S		V_2 = V_{FBX} = 400 V dc					

- - -

See footnote at end of table.

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6.3.2 Diode types 1N1484, 1N1485, and 1N1486 (see figure 2). Diode types 1N3030, 1N3031, and

TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Low temperature operation:		T_A = -65°C					
Reverse blocking current	4211	DC method, condition B		I_{RBX3}	---	5	µA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 kΩ					
2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_2 = 2 kΩ					
2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS 2N2328, S, A, AS 2N2329, S		V_R = 50 V dc V_R = 100 V dc V_R = 200 V dc V_R = 300 V dc V_R = 400 V dc					
Forward blocking current	4206	DC method, condition B		I_{FBX3}	---	5	µA dc
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 kΩ					
2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_2 = 2 kΩ					
2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS 2N2328, S, A, AS 2N2329, S		V_{FBX} = 50 V dc V_{FBX} = 100 V dc V_{FBX} = 200 V dc V_{FBX} = 300 V dc V_{FBX} = 400 V dc					
Gate trigger voltage and current	4221	V_2 = V_{FBX} = 6 V dc, R_L = 100Ω					
2N2323 thru 2N2329 and 2N2323S thru 2N2329S		R_e = 1 kΩ		V_{GT2} I_{GT2}	--- ---	1.0 350	V dc µA dc
2N2323A thru 2N2328A and 2N2323AS thru 2N2328AS		R_e = 2 kΩ		V_{GT2} I_{GT2}	--- ---	0.8 75	V dc µA dc

- - -

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 4							
Exponential rate of voltage rise	4231	T_A = 125°C, condition B, 50Ω ≤ R_L ≤ 400Ω, C = 0.1 to 1.0 µF, repetition rate = 60 pps, test duration = 15 seconds		V_{FBX}			
2N2323 thru 2N2329 and 2N2323S thru 2N2329S		dv/dt = 1.8 v/µs, R_3 = 1 kΩ					
2N2323A thru 2N2328A and 2N2323AS thru 2N2328AS		dv/dt = 0.7 v/µs, R_3 = 2 kΩ					
2N2323, S, A, AS		V_{AA} = 50 V dc			47	---	V dc
2N2324, S, A, AS		V_{AA} = 100 V dc			95	---	V dc
2N2326, S, A, AS		V_{AA} = 200 V dc			190	---	V dc
2N2328, S, A, AS		V_{AA} = 300 V dc			285	---	V dc
2N2329, S		V_{AA} = 400 V dc			380	---	V dc
Forward "on" voltage	4226	i_{FM} = 4a(pk) (pulse), pulse width = 8.5 ms, max; duty cycle = 2% max		V_{FM}	---	2.2	v(pk)
Holding current	4201	Condition B, V_{AA} = 24 V dc max, I_{F1} = 100 mA dc, I_{F2} = 10 mA dc, gate trigger source voltage = 6 V dc, trigger pulse width = 25 µs minimum, R_2 = 330Ω		I_{HDX}	---	2.0	mA dc
2N2323 thru 2N2329 and 2N2323S thru 2N2329S		R_3 = 1 kΩ					
2N2323A thru 2N2328A and 2N2323AS thru 2N2328AS		R_3 = 2 kΩ					
Subgroup 5							
Not applicable							

- - -

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 6							
Surge current	4066	See figure 5; T_A = 122°C, I_{FM} = 15 a, 10 surges, 1 per minute, I_O = 0 at rated V_{RWM}, f = 60 Hz					
Subgroup 7							
Nonrepetitive peak voltage	---	See figure 6, 10 pulses					
2N2323, S, A, AS		V_{RM} = 75 V(pk)					
2N2324, S, A, AS		V_{RM} = 150 V(pk)					
2N2326, S, A, AS		V_{RM} = 300 V(pk)					
2N2328, S, A, AS		V_{RM} = 400 V(pk)					
2N2329, S		V_{RM} = 500 V(pk)					
Endpoint electrical measurements		See table IV, steps 1, 2, 3, and 4					

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification and large lot quality conformance 1/ inspection LTPD	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figure 1		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051			
Hermetic seal a. Fine b. Gross	1071			
Electrical measurements		See table IV, steps 1, 2, 3, and 4		
Decap-internal visual (design verification)	2075			
SEM	2077			
Bond strength	2037			
Die shear	2017			
Subgroup 4				
Intermittent operation life	1037	T_A = 75°C +5°C, I_O = 220 mA dc See figure 4, t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air on the device shall be permitted.		
2N2323 thru 2N2329 and 2N2323S thru 2N2329S		R_{GK} = 1 kΩ		
2N2323A thru 2N2328A and 2N2323AS thru 2N2328AS		R_{GK} = 2 kΩ		
Electrical measurements		See table IV, steps 1, 2, 3, and 4		

- - -

See footnote at end of table.

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Thermal shock | 1051 |

TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance 1/ inspection LTPD	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 5				
Accelerated steady-state operation life	1027	T_A = 175°C or adjusted to give a T_J = 225°C, I_O = 220 mA dc, See figure 4		
Electrical measurements		See table IV, steps 1, 2, 3, and 4		
Bond strength (Al-Au die interconnects only)	2037			
Subgroup 6				
Thermal resistance	3181	$R_{\theta JC}$ = 12.5 °c/W max		

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IIb. Group B inspection for JAN, JANTX and JANTXV.

Inspection	MIL-STD-750		Qualification and large lot quality conformance 1/ inspection LTPD	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 2				
Thermal shock (temperature cycling)	1051			
Hermetic seal	1071			
a. Fine leak b. Gross leak				
Electrical measurements		See table IV, steps 1, 2, 3, and 4		
Subgroup 3				
Forward blocking voltage		See figure 7, T_A = 122°C ±3°C, t = 170 ±24 hours,		
2N2323 thru 2N2329 2N2323S thru 2N2329S		R_{GK} = 1 kΩ		
2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_{GK} = 2 kΩ		
2N2323, S, A, AS 2N2324, S, A, AS 2N2326, S, A, AS 2N2328, S, A, AS 2N2329, S		V_{FBX} = 50 V dc V_{FBX} = 100 V dc V_{FBX} = 200 V dc V_{FBX} = 300 V dc V_{FBX} = 400 V dc		
Electrical measurements		See table IV, steps 1, 2, 3, and 4		
Subgroup 4				
Decap internal visual (design verification)	2075			
Bond strength	2037			

- - -

See footnote at end of table.

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TABLE IIb. Group B inspection for JAN, JANTX and JANTXV - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance 1/ inspection LTPD	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 5				
Thermal resistance	3181	$R_{\theta JC}$ = 12/5 °C/W max		
Subgroup 6				
High-temperature (nonoperating)	1032	T_A = 150°C		
Electrical measurements		See table IV, steps 1, 2, 3, and 4		

1/ For LTPD numbers, see MIL-S-19500.

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TABLE III. Group C inspection.

Inspection	MIL-STD-750		Qualification and large lot quality conformance 1/ inspection LTPD	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 1				
Physical dimensions	2026			
Subgroup 2				
Thermal shock (glass strain)	1056			
Terminal strength (lead fatigue)	2036	Test condition E		
Hermetic seal	1071			
a. Fine leak b. Gross leak				
Moisture resistance	1021			
Electrical measurements		See table IV, steps 1, 2, 3, and 4		
Subgroup 3				
Shock	2016			
Vibration, variable frequency	2056			
Constant acceleration	2006			
Electrical measurements		See table IV, steps 1, 2, 3, and 4		
Subgroup 4				
Salt atmosphere (corrosion)	1041			
Subgroup 5				
Barometric pressure (reduced)	1001	Pressure = 15 mmHg		
2N2328, S, A, AS 2N2329, S				
Measurement during test		See table IV, step 1		

- - -

See footnote at end of table.

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TABLE III. Group C inspection - Continued.

Inspection	MIL-STD-750		Qualification and large lot quality conformance 1/ inspection LTPD	Small lot quality conformance inspection n/c 1/
	Method	Conditions		
Subgroup 6				
Intermittent operation life	1037	See figure 4; t_{on} = 50 ±5 minutes, t_{off} = 10 ±2 minutes, T_A = 75°C ±5°C; I_O = 220 mA dc		
2N2323 thru 2N2329 and 2N2323S thru 2N2329S		R_{GK} = 1 kΩ		
2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_{GK} = 2 kΩ		
Electrical measurements		See table IV, steps 1, 2, 3, and 4		

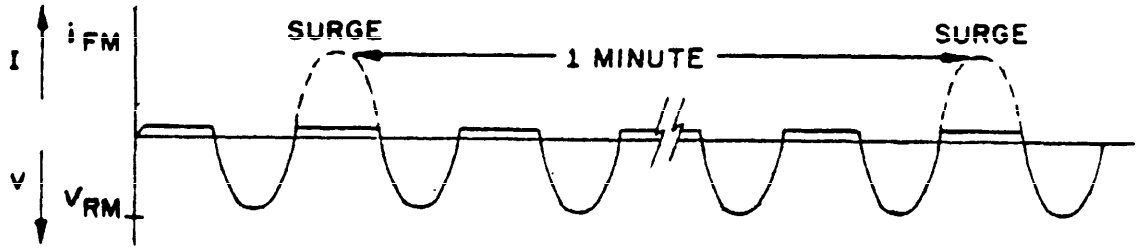
1/ For LTPD numbers, see MIL-S-19500.

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TABLE IV. Groups A, B, and C electrical measurements.

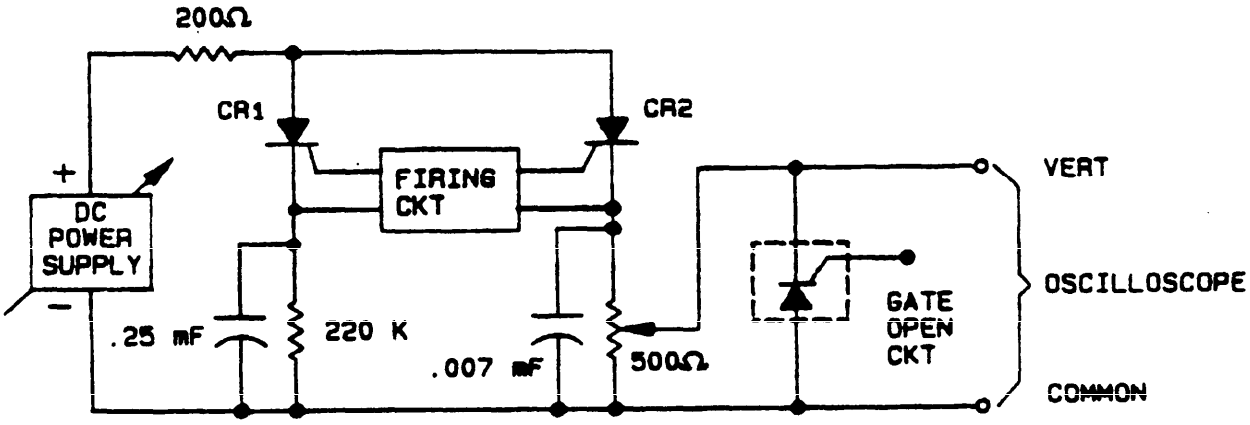
Step	Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
1.	Reverse blocking current	4211	DC method condition B	I_{RBX1}		10	µA dc
	2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 kΩ				
	2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_2 = 2 kΩ				
	2N2323, S, A, AS		V_R = 50 V dc				
	2N2324, S, A, AS		V_R = 100 V dc				
	2N2326, S, A, AS		V_R = 200 V dc				
	2N2328, S, A, AS		V_R = 300 V dc				
	2N2329, S		V_R = 400 V dc				
2.	Forward blocking current	4206	DC method condition B	I_{FBX1}		10	µA dc
	2N2323 thru 2N2329 2N2323S thru 2N2329S		R_2 = 1 kΩ				
	2N2323A thru 2N2328A 2N2323AS thru 2N2328AS		R_2 = 2 kΩ				
	2N2323, S, A, AS		V_{FBX} = 50 V dc				
	2N2324, S, A, AS		V_{FBX} = 100 V dc				
	2N2326, S, A, AS		V_{FBX} = 200 V dc				
	2N2328, S, A, AS		V_{FBX} = 300 V dc				
	2N2329, S		V_{FBX} = 400 V dc				
3.	Forward "on" voltage	4226	i_{FM} = 4 a (pk) (pulse), pulse width = 8.5 ms, max; duty cycle = 2% max	V_{FM}	---	2.2	V(pk)
4.	Gate trigger voltage	4221	$V_2 = V_{FBX}$ = 6 V dc, R_L = 100Ω	V_{FM}			
	2N2323 thru 2N2329 and 2N2323S thru 2N2329S		R_e = 1 kΩ	V_{GT2} I_{GT2}	0.35 ---	0.8 200	V dc µA dc
	2N2323A thru 2N2328A and 2N2323AS thru 2N2328AS		R_e = 2 kΩ	V_{GT2} I_{GT2}	0.35 ---	0.6 20	V dc µA dc

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Procedure: The above waveform shall be applied to the devices under test. During this test, the gate shall be connected to the anode through a suitable firing resistor and series diode or fired by an equivalent method. The device shall be operating at the specified maximum reverse voltage and specified temperature in a single phase circuit with a 60 Hz supply and a resistive load. The total peak device current during the surge shall be 15 amperes.

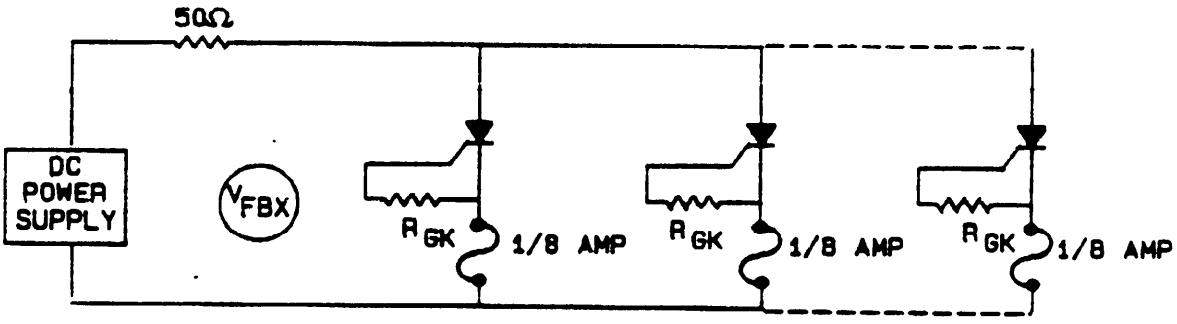
FIGURE 5. Surge current waveform.



Procedure: Adjust potentiometer to achieve the specified voltage accross the device under test. Timing sequence of firing circuit is: Apply trigger pulse to CR2 once every 15 seconds, apply trigger pulse to CR1, 5.0 ±1.0 milliseconds after firing CR2. The specified number of pulses shall be applied. Alternate circuits shall apply rectangular pulses of 5.0 ±1.0 milliseconds with a maximum rise and fall time of 50 microseconds.

FIGURE 6. Nonrepetitive peak reverse voltage test.

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Porcedure: Adjust the power supply to the specified value for the device under test.

NOTES:
1. Rate of rise of voltage must be limited to values specified in group A inspection.
2. One-eight ampere fuse may be replaced by an equivalent load protecting device.

FIGURE 7. Blocking-voltage test.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish may be specified (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Navy - SH
Air Force - 11, 85
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1102)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-19500/276B	Semiconductor Device, Thyristors (Controlled Rectifiers), Silicon Types 2N2323, 2N2324, 2N2328, 2N2329 and S Versions 2N2323A, 2N2324A, 2N2324A, 2N2326A, 2N2328A, and S Versions, JAN, JANTX, JANTXV, and JANS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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